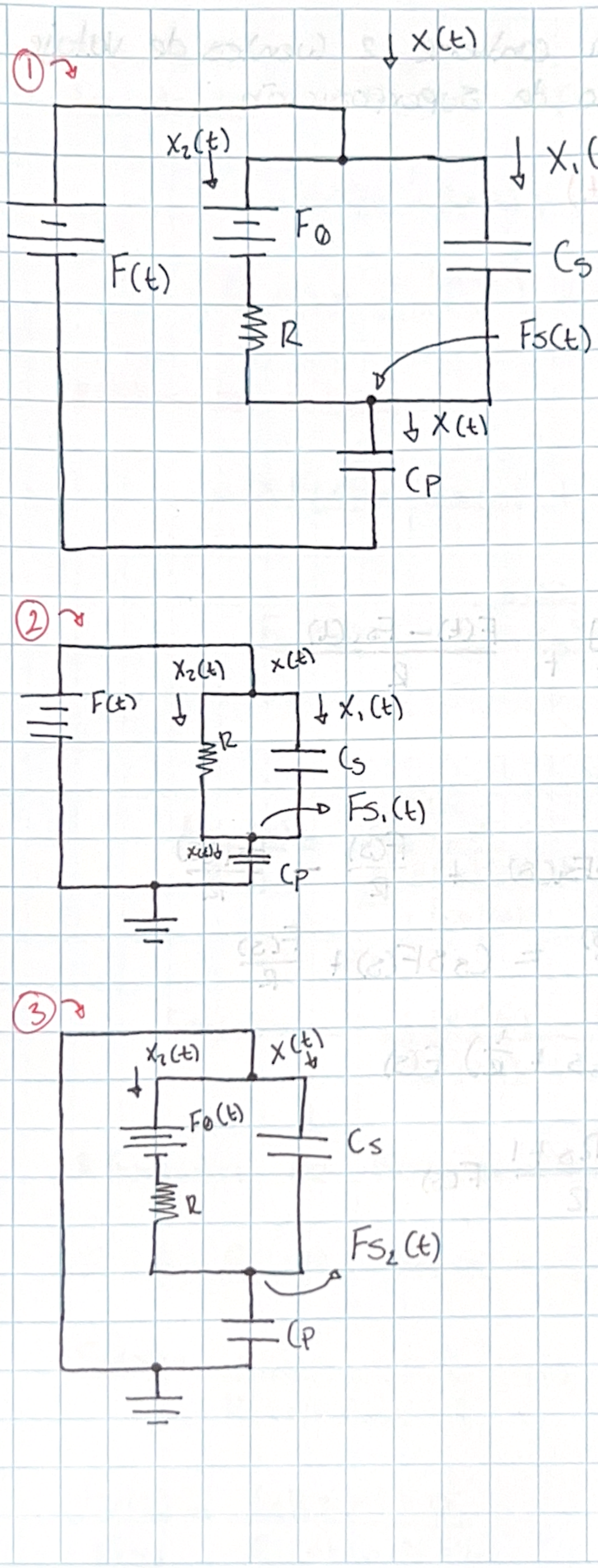
Diagrama

26-Octubre-2026



Análisis Matemático

Debido a que el sistema contiene 2 fuentes de voltaje se debe aplicar el Principio de Superposición

Circuito 1. Apagando $F_0(t)$

$$X(t) = X_1(t) + X_2(t)$$

$$X(t) = C_p \frac{dF_{S1}(t)}{dt}$$

$$X_1(t) = C_S \frac{d[F(t) - F_{S1}(t)]}{dt}$$

$$X_2(t) = \frac{F(t) - F_{S1}(t)}{R}$$

$$C_p \frac{dF_{S1}(t)}{dt} = C_S \frac{d[F(t) - F_{S1}(t)]}{dt} + \frac{F(t) - F_{S1}(t)}{R}$$

*Aplicando Laplace

$$C_p S F_{S1}(S) = C_S S F(S) - C_S S F_{S1}(S) + \frac{F(S)}{R} - \frac{F_{S1}(S)}{R}$$

$$C_p S F_{S1}(S) + C_S S F_{S1}(S) + \frac{F_{S1}(S)}{R} = C_S S F(S) + \frac{F(S)}{R}$$

$$\left(C_p S + C_S S + \frac{1}{R}\right) F_{S1}(S) = \left(C_S S + \frac{1}{R}\right) F(S)$$

$$\frac{R(C_p + C_S)S + 1}{R} F_{S1}(S) = \frac{C_S R S + 1}{R} F(S)$$

$$F_{S1}(S) = \frac{\frac{C_S R S + 1}{R}}{\frac{R(C_p + C_S)S + 1}{R}} F(S)$$

$$F_{S1}(S) = \frac{C_S R S + 1}{R(C_p + C_S)S + 1} F(S)$$

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Circuito 2. Apagando F(t)

Recordando que $F_0(t) = \alpha F(t)$

$$-\alpha F(t) = Rx(t) + \frac{1}{C_p + C_s}\int x(t)\,dt$$

$$F_{S2}(t) = \frac{1}{C_p + C_s}\int x(t)\,dt$$

* Aplicando Laplace

$$-\alpha F(s) = RX(s) + \frac{X(s)}{(C_p + C_s)s}$$

$$F_{S2}(s) = \frac{X(s)}{(C_p + C_s)s}$$

$$\frac{F_{S2}(s)}{F(t)} = \frac{\frac{X(s)}{(C_p + C_s)s}}{-\frac{1}{\alpha}\frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}X(s)} = -\frac{\alpha}{R(C_p + C_s)s + 1}$$

$$F_{S2}(s) = -\frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

* Función de transferencia

$$F_s(s) = F_{s1}(s) + F_{S2}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1}F(s) - \frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

$$F_s(s) = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1}F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1}$$

Error en estado estacionario

$$e(s) = \lim_{s \to 0} s\, F(s) \left(1 - \frac{F_3(s)}{F(s)}\right) = \lim_{s \to 0} s \frac{1}{s} \left(1 - \frac{RC_3 s + 1 - \alpha}{RC_P + RC_3 s + 1}\right) = \alpha$$

Estabilidad en lazo abierto

$$(RC_P + RC_3)s + 1 = 0$$

$$\lambda = \frac{1}{RC_P + RC_3}$$

El sistema es estable.